

# Chapter 10 – Google Workspace Migration

## Learning Objectives

After completing this chapter, you should be able to:

- Understand migration planning.
  - Choose the correct migration strategy.
  - Select the appropriate migration tool.
  - Explain coexistence.
  - Plan enterprise migrations.
  - Troubleshoot migration issues.
  - Answer migration interview questions confidently.
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## 1. What is Migration?

Migration is the process of moving users, data, and services from one platform to another.

Typical migrations include:

- Microsoft Exchange → Google Workspace
- Microsoft 365 → Google Workspace
- IBM Notes → Google Workspace
- IMAP servers → Google Workspace
- Google Workspace Tenant → Another Google Workspace Tenant

Migration is not only about moving email.

Enterprise migrations may also include:

- Mail
  - Calendars
  - Contacts
  - Drive files
  - Shared Drives
  - Permissions
  - User accounts
  - Groups
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## 2. Migration Lifecycle

A successful migration follows a structured lifecycle.

Assessment

↓

Discovery

↓

Planning

↓

Pilot Migration

↓

User Communication

↓

Production Migration

↓

Validation

↓

Support

↓

Project Closure

Skipping planning often leads to migration failures.

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## 3. Assessment Phase

Before selecting a tool,

understand the environment.

Questions to ask:

- What is the source platform?
  - How many users?
  - How much data?
  - Mailbox sizes?
  - Shared mailboxes?
  - Distribution groups?
  - Public folders?
  - Bandwidth?
  - Downtime requirements?
  - Compliance requirements?
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## Example

Company

Employees

15,000

Source

Microsoft Exchange 2019

Mailbox Size

40 TB

Requirement

Less than 2 hours downtime

This information determines the migration approach.

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## 4. Migration Strategies

### Cutover Migration

All users migrate at once.

Example

Friday evening

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Migrate all users

↓

Update MX records

↓

Monday morning

↓

Everyone uses Google Workspace

Advantages

- Simple
- Short project duration
- One transition event

Disadvantages

- Higher risk
- Requires careful planning
- Difficult to roll back

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## Phased Migration

Users migrate in batches.

Example

Week 1

HR

Week 2

Finance

Week 3

Engineering

Week 4

Sales

Advantages

- Lower risk
- Easier troubleshooting
- Smaller impact
- Better user support

Disadvantages

- Longer project
- Temporary coexistence

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## Coexistence

Both platforms operate simultaneously.

Example

Some users remain on Exchange.

Others use Google Workspace.

Mail continues flowing between both environments.

Used when:

- Large enterprises
  - Regulatory constraints
  - Complex migrations
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## Strategy Comparison

| Strategy    | Best For                      | Risk                    | Duration |
|-------------|-------------------------------|-------------------------|----------|
| Cutover     | Small organizations           | High                    | Short    |
| Phased      | Medium to large organizations | Low                     | Medium   |
| Coexistence | Large enterprises             | Lowest operational risk | Long     |

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## Interview Scenario

Management says:

"We want a cutover migration."

Your response:

"A cutover migration is possible if the business accepts the associated downtime and risk. For a large organization, I would recommend a phased migration over approximately three weeks to reduce operational risk, validate each batch, and provide better user support."

This demonstrates both technical understanding and business judgment.

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## 5. Google Migration Tools

Google provides several migration tools.

Each is designed for a different use case.

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### Data Migration Service (DMS)

Purpose

Simple mailbox migration.

Supports:

- IMAP
- Microsoft Exchange

- Microsoft 365
- Google Workspace (supported scenarios)

Migrates:

- Email
- Contacts
- Calendars

Advantages

- Built into the Admin Console
- Easy to configure
- No additional server required

Limitations

- Does not migrate all enterprise workloads
- Limited compared to larger migration solutions

Best for:

Small and medium-sized migrations.

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## **Google Workspace Migration for Microsoft Exchange (GWMME)**

Purpose

Enterprise Exchange migrations.

Migrates:

- Mail
- Calendar
- Contacts

Supports:

- Microsoft Exchange

Advantages

- Enterprise-grade
- Highly configurable
- Supports staged migrations
- Handles large Exchange environments

Best for:

Large Exchange deployments.

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## Google Workspace Migration for Microsoft Outlook (GWMMO)

Purpose

Migrate Outlook profile data.

Migrates:

- PST files
- Email
- Calendar
- Contacts
- Notes (where supported)

Best for:

Individual users or small groups.

Example

A user has years of archived email stored in a local PST file.

GWMMO imports the PST into Gmail.

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## Google Workspace Migrate

Purpose

Enterprise content migration.

Supports migration of:

- File servers
- SharePoint
- OneDrive
- Shared Drives
- Google Drive
- Collaboration content (supported workloads)



## Advantages

- Enterprise scale
- Detailed reporting
- Incremental migrations
- Permission preservation (where supported)

Best for:

Large organizations migrating collaboration content.

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## CloudM

CloudM is a third-party migration platform.

It supports:

- Google Workspace
- Microsoft 365
- Exchange
- Tenant-to-tenant migrations
- Drive migrations
- User provisioning
- Archive migrations

Think of CloudM as an enterprise migration suite.

Unlike Google's individual tools,

CloudM provides a unified migration platform with additional automation and reporting capabilities.

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## Tool Comparison

| Tool                     | Best Use Case                                             |
|--------------------------|-----------------------------------------------------------|
| Data Migration Service   | Small mailbox migrations                                  |
| GWMME                    | Microsoft Exchange migrations                             |
| GWMMO                    | Outlook/PST migrations                                    |
| Google Workspace Migrate | Enterprise file and collaboration content migration       |
| CloudM                   | Complex enterprise migrations and cross-platform projects |

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# Example Scenarios

## Scenario 1

Source

Exchange

20,000 users

Recommended Tool

✓ GWMME

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## Scenario 2

Source

Outlook PST

200 users

Recommended Tool

✓ GWMMO

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## Scenario 3

Source

Small IMAP server

100 users

Recommended Tool

✓ Data Migration Service

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## Scenario 4

Source

SharePoint

OneDrive

File Servers

Recommended Tool

✓ Google Workspace Migrate

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## Scenario 5

Complex merger

Microsoft 365

Google Workspace

Cross-tenant migration

Recommended Tool

✓ CloudM

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## 6. Pilot Migration

Never migrate everyone first.

Instead,

select a pilot group.

Typical pilot users:

- IT team
- Help Desk
- Department representatives
- Power users

Validate:

- Email
- Calendar

- Contacts
- Permissions
- Mobile devices
- Third-party integrations

Resolve issues before the production migration.

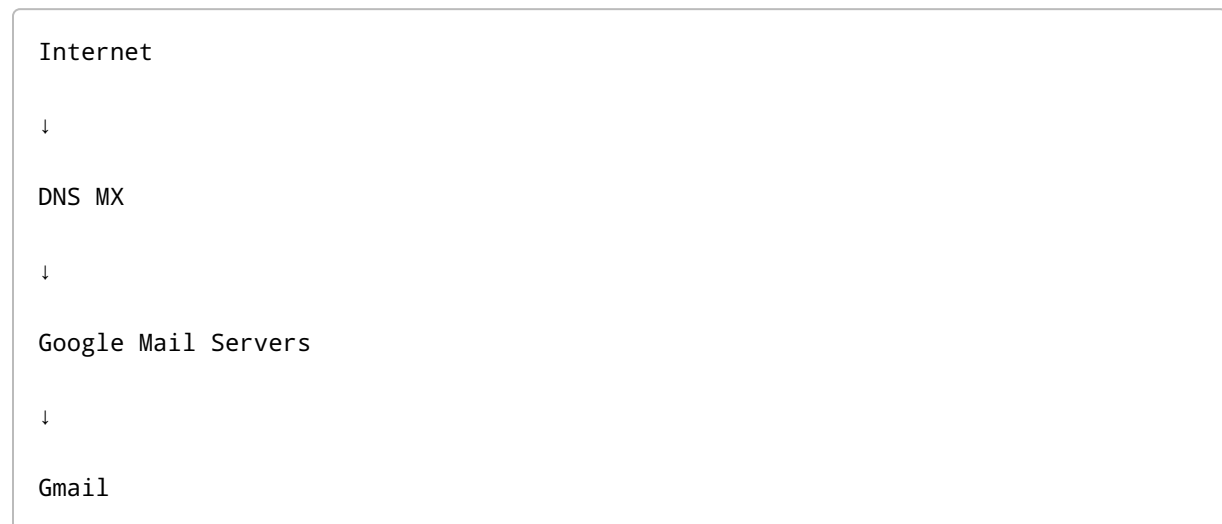
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## 7. MX Cutover

After validating migrated users,

update the MX records.

Flow



Once the MX records propagate,

new inbound mail is delivered to Google Workspace.

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## 8. Migration Validation

After migration,

verify:

- Mailbox count
- Message count
- Calendar events

- Contacts
- Delegates
- Groups
- Aliases
- Mobile access
- Shared Drives (if applicable)
- Permissions

Migration is not complete until validation succeeds.

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## 9. Common Migration Issues

### Missing Email

Possible causes:

- Incomplete migration
  - Sync interrupted
  - Mail filtering
  - Incorrect source selection
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### Calendar Missing

Possible causes:

- Unsupported item
  - Migration skipped
  - User permissions
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### Authentication Failure

Possible causes:

- Incorrect credentials
  - OAuth issues
  - Service account configuration
  - Domain-Wide Delegation (where applicable)
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## Mail Flow Failure After Migration

Symptoms:

Users can send mail.

Users cannot receive external mail.

Investigation:

Check MX records first.

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## Permission Problems

Possible causes:

- Shared Drive permissions not migrated
  - Incorrect ownership mapping
  - External sharing restrictions
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## 10. Enterprise Migration Best Practices

- ✓ Perform a detailed assessment.
  - ✓ Communicate with users early.
  - ✓ Run pilot migrations.
  - ✓ Validate migrated data.
  - ✓ Plan MX cutover carefully.
  - ✓ Keep rollback plans.
  - ✓ Monitor migration logs.
  - ✓ Provide post-migration support.
  - ✓ Document every migration phase.
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# Common Mistakes

- ✗ Skipping the assessment phase.
  - ✗ Migrating without a pilot.
  - ✗ Selecting the wrong migration tool.
  - ✗ Ignoring mailbox size limitations.
  - ✗ Forgetting user communication.
  - ✗ Updating MX records before validation.
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## Real Enterprise Scenario

Company

15,000 users

Source

Microsoft Exchange

Requirement

Minimal disruption.

Recommended approach:

1. Assessment.
2. Pilot with IT.
3. Phased migration over several weeks.
4. Validate each batch.
5. Update MX records after successful validation.
6. Provide post-migration support.

Although a cutover is technically possible,

a phased approach significantly reduces business risk.

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# Interview Questions

1. What are the phases of a migration project?
  2. What is the difference between cutover and phased migration?
  3. When would you recommend coexistence?
  4. Which Google tool would you choose for an Exchange migration?
  5. What is GWMMO used for?
  6. When would you use Data Migration Service?
  7. How does Google Workspace Migrate differ from GWMME?
  8. Where does CloudM fit into a migration strategy?
  9. Why should you always perform a pilot migration?
  10. Users can send email after migration but cannot receive external email. Where do you investigate first?
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## Key Takeaways

Migration is a structured project, not just a data copy.

Choose the migration strategy based on business requirements, downtime tolerance, and risk.

Select the migration tool based on the source platform and the type of data being migrated.

Always perform pilot migrations and validate results before production cutover.

Plan DNS and MX changes carefully to minimize email disruption.

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## Chapter Summary

In this chapter you learned:

✓ Migration lifecycle.



- ✓ Assessment and planning.
- ✓ Cutover, phased, and coexistence strategies.
- ✓ Data Migration Service (DMS).
- ✓ Google Workspace Migration for Microsoft Exchange (GWMME).
- ✓ Google Workspace Migration for Microsoft Outlook (GWMMO).
- ✓ Google Workspace Migrate.
- ✓ CloudM.
- ✓ Migration validation.
- ✓ Enterprise migration best practices.
- ✓ Common migration troubleshooting.

The next chapter will cover **Google Drive, Shared Drives, and Collaboration**, including ownership, permissions, roles, external sharing, Shared Drives vs My Drive, data governance, and enterprise collaboration design—the next major topic we've already started discussing together.